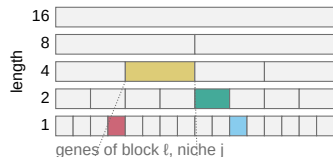


A

1 Index dyadic intervals

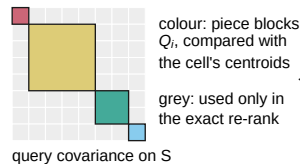


2 Split the query into dyadic pieces



Each run of consecutive genes (in the block's order) gives at most $2 \log_2(\text{run length})$ pieces. Repeated in every niche's gene order.

3 Score each piece



$$d_i(t) = \|\log Q_i - C_i(t)\|_F / \sqrt{p_i}$$

$C_i(t)$: centroid of tile t 's cluster in the cell of piece i ; p_i genes

4 Combine, re-rank

